$$\|x_i\|_2 = 1, \quad d_{n,i} = \min_{j \neq i} \|x_i - x_j\|_2$$

$$\mathcal{L}_{\text{KoLeo}} = -\frac{1}{n} \sum_{i=1}^{n} \log d_{n,i}$$

$$-\log 0.1 \approx 2.303, \quad -\log 0.5 \approx 0.693$$